

Do High Schools Shape College Choices? Evidence from Centralized Assignment in Chile

Bruno E. Aravena-Magüida

May 2026

Abstract

Education is one of the central investments through which societies build human capital, yet we know less about whether high schools shape students' later choices beyond raising test scores. This paper studies that question in Chile, where a centralized school assignment system creates lottery variation in high school offers. I link assignment records to administrative data on school enrollment, standardized achievement, and higher-education enrollment, and ask whether high schools causally affect both achievement and college choices. The empirical strategy proceeds in two steps. First, I estimate observational school value added using rich pre-treatment controls, including grade-4 test scores, family background, and middle-school histories. Second, I use lottery-generated offers to test whether attending schools with higher observational value added improves later outcomes, controlling for the expected value added implied by each student's assignment risk. The results suggest that observational value added contains substantial causal signal. Achievement value added passes through strongly to math and language scores, while higher-education value added also predicts later STEM enrollment and institution quality. An orthogonal value-added exercise shows that the component of STEM value added unrelated to math value added predicts STEM enrollment but not math achievement. These findings suggest that high schools differ not only in their ability to raise test scores, but also in dimensions that shape students' transition into higher education.

1 Introduction

School quality is usually summarized by achievement. This is natural: test scores are widely available, comparable across schools, and predictive of later outcomes. But school and teacher effects are not necessarily one-dimensional. Recent work shows that schools and teachers can affect attendance, behavior, crime, teenage motherhood, labor-market outcomes, and other long-run outcomes in ways that are only weakly related to test-score effects (Jackson 2018; Beuermann et al. 2023; Rose et al. 2022; Angrist et al. 2023). Families and policy-makers might therefore care about school quality measures that capture more than achievement.

This broader view of school quality is especially relevant for high schools. High schools are the last formal educational institutions students attend before making higher-education choices, and those choices are consequential. Whether students enter higher education, which institutions they attend, and which fields they choose all matter because returns vary substantially across institutions and fields of study (Kirkeboen et al. 2016). Yet many measures of school effects on higher education are coarse, focusing on enrollment or persistence rather than on the type of institution or field students enter.

There are also good reasons to think that higher-education choices are shaped by frictions that schools may influence. Prior work documents large differences in application and enrollment behavior across groups of students, and shows that relatively small interventions can change college search, application, and major choice (Hoxby and Turner 2015; Dynarski et al. 2021; Hastings et al. 2016; Chetty et al. 2025). If high schools differ in counseling, information provision, expectations, peer norms, academic confidence, or preparation for particular fields, then their effects on higher education may extend beyond their effects on achievement.

This paper asks whether high schools causally affect higher-education choices, and whether those effects are captured by traditional achievement value added. The empirical challenge is that students and families sort across schools. Families choose schools based on resources, preferences, networks, location, and expectations; these same characteristics may also shape later higher-education choices. As a result, differences across schools may reflect both causal school effects and the underlying characteristics of the students and families who select into them.

I address this challenge using Chile’s centralized school assignment system, the Sistema de Admision Escolar (SAE). SAE assigns students to public and private-subsidized schools through a centralized mechanism that uses random tie-breakers when schools are oversubscribed. The system covers a large national school sector, allowing me to study school effects at a scale that is unusual in lottery-based designs. I combine this assignment variation with administrative data linking school enrollment, standardized tests, and higher-education enrollment.

The analysis proceeds in two steps. First, I estimate observational value added for each school and outcome in the broad student panel. These measures are school fixed effects from regressions that adjust for rich pre-treatment controls, including grade-4 SIMCE

scores, family-background variables, and middle-school controls. Second, I use SAE lottery offers to ask whether schools that look better in observational value-added terms also produce larger causal effects. For each outcome, I compare the value added of the school a student attends with the value added of the school the student is offered, while controlling for the expected value added implied by the student’s assignment probabilities.

I summarize this relationship with a pass-through coefficient, θ . If $\theta = 1$, then a lottery-induced increase in observational value added translates one-for-one into a causal gain in the corresponding outcome. If $\theta = 0$, the observational value-added measure has no causal predictive content along the lottery-induced margin. The main estimates suggest substantial pass-through. For achievement, the estimates are 0.841 for math and 0.835 for language. For higher-education outcomes, the pass-through is 0.674 for STEM enrollment and 0.680 for institution quality, measured using institutional accreditation years. In practical terms, schools that observationally look better at raising achievement or higher-education outcomes also tend to have causal effects in those same outcomes, though the causal effects are smaller than the observational differences.

I then ask for whom observational value-added measures are most informative. Splitting the sample by baseline grade-4 math achievement shows that achievement value added is more predictive of causal effects for higher-achieving students, while the higher-education value-added measures are more stable across the baseline achievement distribution. This does not imply that schools matter only for high achievers; rather, it suggests that the usefulness of a given value-added metric depends on the population to which it is applied.

Finally, I examine whether STEM effects are simply achievement effects in another form. To do this, I construct an orthogonal value-added measure that removes from STEM-enrollment value added the part predicted by math value added. Math and STEM value added are already weakly related, and the IV results confirm that the orthogonal STEM component predicts STEM enrollment but not math achievement. Conversely, math value added predicts math achievement but not STEM enrollment. These results suggest that high schools differ along multiple dimensions, and that value added should not be treated as a single-dimensional measure of school quality.

The paper contributes to three literatures. First, it contributes to work on teacher and school value added. A large literature estimates the effects of teachers and schools on test scores and longer-run outcomes (Chetty et al. 2014; Dobbie and Fryer 2015; Jackson 2018; Beuermann et al. 2023; Rose et al. 2022). This work increasingly emphasizes that test-score value added may miss important dimensions of educational production. I extend this idea to high-school effects on higher-education choices, where the relevant outcomes include field and institution quality rather than only enrollment or achievement.

Second, the paper contributes to the literature using lotteries and centralized assignment systems to estimate school effects (Abdulkadiroğlu et al. 2014; Abdulkadiroğlu et al. 2017; Angrist et al. 2024). In the spirit of Angrist et al. (2017), I ask whether observational value-added measures are useful guides to causal school effects. The contribution here is to apply this pass-through logic to a national high-school assignment system and to outcomes observed after students transition into higher education.

Third, the paper contributes to work on higher-education choice. Prior research shows that the field and institution students choose matter for later outcomes, and that information and application frictions can change those choices (Kirkeboen et al. 2016; Hoxby and Turner 2015; Dynarski et al. 2021; Hastings et al. 2016; Chetty et al. 2025). I study high schools as a possible upstream source of these differences. Rather than evaluating a particular counseling or information intervention, the paper measures whether the high-school system itself generates causal differences in students' higher-education paths.

The rest of the paper is organized as follows. Section 2 describes the Chilean setting and the linked administrative data. Section 3 presents the empirical framework. Section 4 reports the main results, heterogeneity by baseline achievement, and the orthogonal math/STEM value-added exercise. Section 5 concludes.

2 Context and Data

2.1 Institutional Setting

Chile is a useful setting for this project because both school enrollment and higher-education choices are recorded in national administrative data. Formal secondary education begins in grade 9 and lasts four years. Students may attend public, private-subsidized, or private non-subsidized schools. Public and private-subsidized schools are publicly financed and participate in the centralized school assignment system described below; private non-subsidized schools remain outside that assignment mechanism. In the broad grade-8 universe used in the analysis, private non-subsidized schools account for about 9.3% of students. The lottery-based analysis is therefore drawn from the public and private-subsidized sector, with additional sample restrictions for timely SAE application and assignment risk described below.

I construct a student panel that follows students across schools and years using a scrambled individual identifier. The panel records annual enrollment, school identifiers, grade progression, attendance, and grades. I use these histories to define the main high-school exposure as the school identifier (RBD) where a student spends the most time after grade 8. This definition smooths over short enrollment spells and transfers, and it gives a single high-school measure for each student. The same panel also lets me construct pre-treatment middle-school controls, including the school most attended in grades 5 to 8 and standardized measures of grades and attendance before high school.

2.2 Student Panel and Baseline Controls

The broad analysis panel starts from the universe of students first observed in grade 8 between 2017 and 2020. This produces 944,360 students across four cohorts. The panel contains 238,467 students in the 2017 grade-8 cohort, 232,498 in 2018, 233,401 in 2019, and 239,994 in 2020. Students can be followed over time with a scrambled individual identifier, and the main high-school exposure is observed for 4,133 RBDs.

Baseline achievement comes from SIMCE, Chile’s national standardized assessment system. SIMCE is administered in different grades depending on the year, including grades 4, 8, and 10. Because the COVID-19 pandemic disrupted testing in 2020 and 2021, grade-8 and grade-10 measures are not available for all cohorts in the same way. I therefore use grade-4 SIMCE as the main pre-treatment achievement control. This gives a nationally comparable measure of math and language skills before high-school assignment for 771,968 students in the broad panel.

The SIMCE files also contain parent-survey information. I use these surveys to measure family background before treatment, including household income, parental education, indigenous origin of the parents, and early-childhood attendance. These variables are useful because they proxy for family resources and prior investments that may influence both high-school choices and later higher-education outcomes. When parent-survey variables

are missing, I use pre-treatment school and neighborhood cells to impute medians where enough donor observations are available; remaining missing values are left missing.

Table 1: Main data sources and sample sizes

Sample or outcome availability	Students
Broad grade-8 universe, 2017–2020	944,360
With grade-4 SIMCE math and language controls	771,968
With valid higher-education math admission-test outcome	640,078
SAE lottery sample with assignment risk	163,600
SAE lottery sample with math outcome	103,941

2.3 Centralized School Assignment

Chile’s centralized school assignment system, the Sistema de Admision Escolar (SAE), assigns students to participating schools through a version of student-proposing Deferred Acceptance. Families submit ranked lists of schools. The mechanism combines these rankings with school vacancies and priority rules, and it uses random tie-breakers when there are more applicants than seats within a priority group. Conditional on the information that determines assignment probabilities, these tie-breakers generate the quasi-experimental variation used in the empirical design.

I use applications to grade-9 vacancies to construct the lottery sample. The estimation sample consists of timely SAE applicants with assignment risk, meaning that no single school is assigned with probability one in the simulated assignment data. This sample contains 163,600 students from the 2018–2020 cohorts. On average, these students have positive simulated assignment probability at about 3.1 schools; the median is 3 and the 90th percentile is 5. About 86.5% receive a nonzero first-round offer. The first-round offered RBD is the source of the instrument, while the student’s realized high-school exposure is measured by the most-time RBD described above.

The value-added exercise uses the broader student panel, while the causal estimates use the smaller lottery-risk sample. This distinction is important. The broad panel improves precision in estimating school-level outcome gradients. The lottery sample then asks whether lottery-induced movement toward schools with higher measured value added causes improvements in the same outcomes.

2.4 Higher-Education Outcomes

The higher-education system also generates administrative data at the student-program level. Students applying to selective higher education take a national entrance exam: the PSU through 2020, the PTU in 2021–2022, and the PAES from 2023 onward. The mandatory components measure math and verbal skills. Students may also take optional exams in history and social sciences, natural sciences, and advanced mathematics. I use

the mandatory math and language scores as achievement outcomes, standardizing them within test year to make them comparable across test regimes.

Higher-education applications and admissions are organized at the program-institution level. A student's admission depends on entrance-exam scores, school GPA-based scores, ranked program preferences, and the weights each program assigns to score components. This structure is useful because it makes higher-education choices much richer than a single college-enrollment indicator: students choose not only whether to enroll, but also which institution, program, and field to pursue.

I link the school panel to higher-education matricula records for first observed enrollment. The main non-achievement outcome is an indicator for enrollment in a STEM program, defined from the field classification of the first higher-education enrollment. I also construct measures of program and institution accreditation. In the main results, I focus on institution quality measured by the number of years for which the institution is accredited, without interacting this measure with program accreditation. These outcomes allow me to study whether high schools affect the type and quality of higher-education option students enter, not just whether students enter higher education at all.

Students without an observed higher-education enrollment are coded as zero for unconditional enrollment outcomes such as STEM enrollment. For accreditation-years outcomes, non-enrollment is also coded as zero, while observed enrollment records with missing accreditation inputs are left missing. In the broad panel, 626,413 students have a first higher-education enrollment record, and the unconditional STEM enrollment rate is 20.9%.

2.5 School Value-Added Measures

For each outcome, I construct observational school value added using the broad student panel. The adjusted value-added measure is the school fixed effect from a student-level regression with cohort, demographic, baseline SIMCE, family-background, and middle-school controls. The All-sample value-added file contains 4,068 schools and 42 outcome transformations. Controlled math value added is available for 3,365 schools, controlled language value added for 3,374 schools, and controlled STEM and accreditation value added for 3,685 schools.

These school measures are not interpreted as causal by themselves. They summarize how students linked to a school perform after adjusting for rich pre-treatment information. The causal analysis uses the SAE lotteries to ask whether being offered and then attending schools with higher observational value added improves student outcomes.

3 Empirical Framework

This section describes the empirical object and the research design. The goal is not to estimate a separate causal effect for every school and every possible margin of substitution. Instead, I ask whether an outcome-specific observational value-added measure is informative about causal effects generated by lottery-induced changes in school attendance. The design therefore collapses school effects into scalar measures and estimates a pass-through coefficient for each outcome.

3.1 School Effects and Observational Value Added

Let $Y_i^m(j)$ denote student i 's potential outcome for outcome m if assigned to high school j . I use a constant-effects value-added model as a guide:

$$Y_i^m(j) = \mu^m + V_j^m + \epsilon_i^m, \quad (1)$$

where V_j^m is the contribution of school j to outcome m , measured relative to an average school. The main outcomes are achievement in math and language, STEM enrollment, and institution quality in higher education.

The first step constructs observational value added in the broad grade-8 panel. For each outcome m , I estimate

$$Y_i^m = \alpha^m + \lambda_{s(i)}^m + X_i' \Gamma^m + u_i^m, \quad (2)$$

where $s(i)$ is the student's main high-school RBD and X_i contains pre-treatment controls. These include cohort, gender, age, grade-4 SIMCE math and language, family-background measures from SIMCE parent surveys, and middle-school controls from grades 5 to 8. The observational value-added measure $\hat{V}_j^m$ is the centered school fixed effect $\hat{\lambda}_j^m$ from equation 2.

These fixed effects are not interpreted as causal by themselves. They combine true school effects with any remaining selection not absorbed by controls. The second step asks whether the part of $\hat{V}_j^m$ shifted by the SAE lotteries corresponds to causal effects in the same outcome.

3.2 Assignment Risk and Expected Value Added

SAE applicants submit rank-ordered lists of schools. Let Z_i denote the first-round school offer and D_i denote the school attended, measured by the post-grade-8 most-time RBD. A student has a preference and priority type θ_i , and the assignment mechanism implies an assignment-probability vector $p_i = (p_{i1}, \dots, p_{iJ})$, where $p_{ij} = \Pr(Z_i = j)$. Under equal treatment of equals, students with the same type have the same assignment probabilities (Abdulkadiroğlu et al. 2017). For pre-treatment characteristics and potential outcomes fixed under re-randomization,

$$(Y_i^m(1), \dots, Y_i^m(J), X_i) \perp\!\!\!\perp Z_i \mid p_i. \quad (3)$$

The raw vector p_i is high-dimensional, because students can rank many different combinations of schools. For an outcome-specific scalar value-added measure, I summarize assignment risk by the expected value added in the student’s portfolio:

$$E_i^m = \sum_j p_{ij} \hat{V}_j^m. \quad (4)$$

This is the average value added the student would receive over repeated randomizations of the assignment mechanism, holding the submitted application and priorities fixed. In the main IV specification, this scalar expected value replaces a large set of school-specific probability controls.

For each outcome m , define the value added of the attended school and offered school as

$$A_i^m = \hat{V}_{D_i}^m, \quad O_i^m = \hat{V}_{Z_i}^m. \quad (5)$$

The instrument is O_i^m , the value added of the school generated by the lottery offer. The endogenous variable is A_i^m , the value added of the school actually attended. Because assignment offers vary randomly conditional on the assignment probabilities, variation in O_i^m around E_i^m provides quasi-experimental variation in the quality of the school offer.

3.3 Pass-Through IV Specification

The main estimating equation is

$$Y_i^m = \alpha^m + \theta^m A_i^m + \rho^m E_i^m + X_i' \delta^m + \eta_i^m, \quad (6)$$

where A_i^m is instrumented with O_i^m . Equivalently, the first stage is

$$A_i^m = a^m + \kappa^m O_i^m + \rho_A^m E_i^m + X_i' \delta_A^m + \nu_i^m. \quad (7)$$

All reported specifications use heteroskedasticity-robust standard errors.

The coefficient θ^m is the key object. It measures how much of the observational value-added difference induced by the lottery offer passes through into causal effects on the same outcome. If $\theta^m = 1$, then a one-unit increase in observational value added causes a one-unit increase in the outcome. If $\theta^m = 0$, the observational measure has no causal predictive content along the lottery-induced margin. Values between zero and one imply that observational value added is directionally informative, but overstates the causal effect of attending higher-value-added schools.

This approach differs from estimating a full vector of school-specific causal effects. With many schools and many possible fallback options, individual school instruments can mix several margins of substitution, making school-specific IV effects difficult to interpret (Angrist et al. 2023; Kirkeboen et al. 2016). The scalar design instead asks a simpler question: whether movement toward schools with higher measured value added causes better outcomes, after controlling for the expected value added implied by each student’s assignment risk.

3.4 Heterogeneity and Orthogonal Value Added

I use two extensions of equation ???. First, I estimate the same pass-through coefficient separately by quintiles of baseline grade-4 math achievement. Because grade-4 SIMCE is measured before high-school assignment, these splits describe heterogeneity by pre-treatment achievement without conditioning on a post-treatment outcome.

Second, I construct two-dimensional value added for math achievement and STEM enrollment. The purpose is to ask whether STEM value added is only another expression of math value added, or whether it contains an independent school dimension. Let $\hat{V}_j^M$ be math value added and $\hat{V}_j^S$ be STEM-enrollment value added. I residualize STEM value added on math value added:

$$\hat{V}_j^{S\perp M} = \hat{V}_j^S - \hat{\phi}\hat{V}_j^M. \quad (8)$$

I then estimate a two-regressor IV model in which attended math value added and attended orthogonal STEM value added are both endogenous, and their offered analogues are the instruments. For outcome $m \in \{M, S\}$:

$$Y_i^m = \alpha^m + \theta_M^m A_i^M + \theta_{S\perp M}^m A_i^{S\perp M} + \rho_M^m E_i^M + \rho_{S\perp M}^m E_i^{S\perp M} + X_i' \delta^m + \eta_i^m. \quad (9)$$

This exercise follows the logic of orthogonal value-added measures used to distinguish achievement and non-achievement dimensions of educational production (Jackson 2018; Rose et al. 2022). In this setting, it tests whether schools with high STEM-enrollment value added are simply high math-value-added schools, or whether they have a distinct effect on students' higher-education field choices.

4 Results

I organize the results in three blocks. First, I report the main pass-through estimates for four scalar value-added measures: two achievement measures and two higher-education measures. Second, I ask whether these pass-through estimates vary by baseline achievement, using grade-4 math achievement quintiles. Third, I use the two-dimensional value-added construction to separate math value added from the component of STEM value added that is orthogonal to math value added.

4.1 Main Value-Added Estimates

Table ?? reports the main IV estimates. Each column uses the same outcome to construct the observational school value-added measure, the lottery-induced expected value-added instrument, and the realized attended school value-added treatment. The coefficients can therefore be read as pass-through estimates: how much a lottery-induced increase in a school’s observational value added translates into a causal gain in the corresponding student outcome.

The achievement estimates are close to one. A one standard deviation increase in school math value added raises math achievement by 0.841 standard deviations, while the analogous estimate for language is 0.835. The higher-education estimates are also positive, though somewhat smaller and less precise. The estimated pass-through is 0.674 for STEM enrollment and 0.680 for institution quality, measured using institutional accreditation years without interacting it with program accreditation. Taken together, the estimates suggest that the observational school value-added measures are informative not only about test score production, but also about later higher-education choices.

Table 2: Main scalar school-value IV estimates

	Achievement		Higher education	
	Math	Language	STEM enrollment	Institution quality
θ	0.841	0.835	0.674	0.680
SE	(0.071)	(0.082)	(0.065)	(0.095)
Observations	90,906	92,472	135,898	135,898

4.2 Heterogeneity by Baseline Achievement

Figure ?? repeats the same four estimates separately by quintiles of baseline grade-4 math achievement. This comparison is useful because it keeps the heterogeneity split pre-treatment and available for the full analysis sample.

The achievement pass-through estimates increase with baseline math achievement, especially for math value added. In the bottom quintile, the math estimate is small and imprecise, while in the top quintile it is above one. The language estimates follow a

similar, though less monotone, pattern. In contrast, the STEM enrollment estimates are stable across the baseline achievement distribution, with point estimates between 0.64 and 0.73 in all quintiles. The institution-quality estimates are positive in all quintiles and are generally below the unit-pass-through benchmark. This pattern suggests that the test-score value-added measures may be more predictive for higher-achieving baseline students, while the higher-education value-added measures appear to travel more evenly across baseline achievement groups.

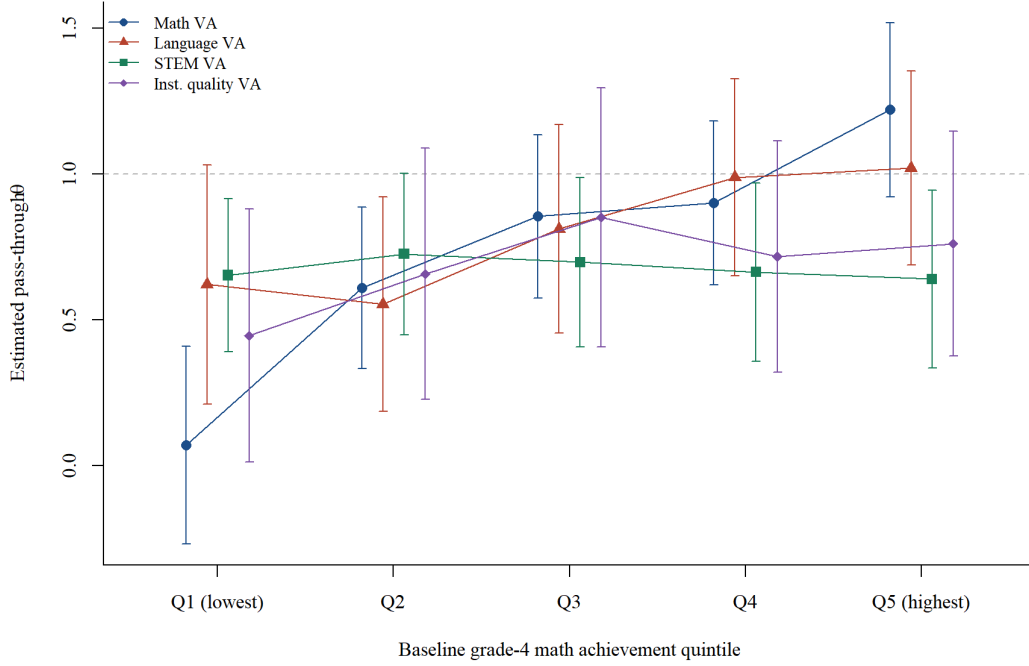


Figure 1: Scalar school-value IV estimates by baseline grade-4 math achievement quintile

4.3 Orthogonal Math and STEM Value Added

The scalar estimates above treat each outcome-specific value-added measure one at a time. I also construct a two-dimensional measure that separates math value added from STEM-enrollment value added. Figure ?? shows that the raw school-level relationship between math value added and STEM enrollment value added is positive but weak. This makes it possible to ask whether schools that improve STEM enrollment are doing so only because they are also high math value-added schools, or whether there is a distinct STEM-related component.

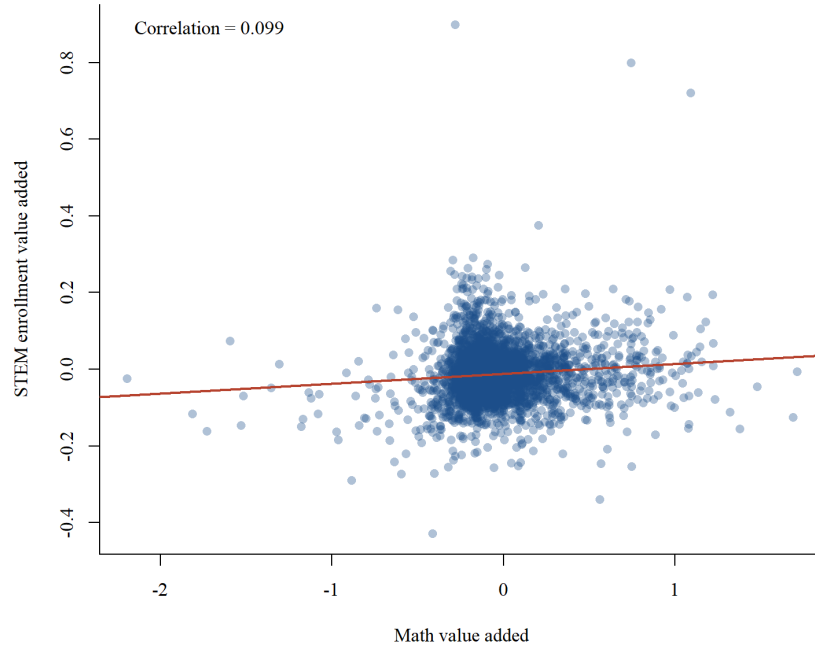


Figure 2: Relationship between math and STEM observational value added

Table ?? reports the corresponding two-regressor IV estimates. Math value added strongly predicts math achievement, with an estimate of 0.844, but it does not predict STEM enrollment once the orthogonal STEM component is included. Conversely, the STEM component orthogonalized against math value added strongly predicts STEM enrollment, with an estimate of 0.656, while it has essentially no relationship with math achievement. This supports the interpretation that schools differ along at least two dimensions: one dimension that raises measured achievement, and another that raises STEM enrollment conditional on achievement value added.

Table 3: Two-dimensional orthogonal school-value IV estimates

	Math score	STEM enrollment
Math VA	0.844 (0.072)	-0.058 (0.039)
STEM VA orthogonal to math	0.017 (0.111)	0.656 (0.066)

5 Conclusion

This paper studies whether high schools affect higher-education choices in ways that are captured by, or distinct from, traditional achievement value added. Using Chile’s centralized school assignment system and linked administrative data, I first estimate observational school value added in a broad national panel and then use SAE lottery offers to test whether those measures predict causal effects.

The results suggest that observational value added contains substantial causal signal. Achievement value added passes through strongly to achievement: the estimates are 0.841 for math and 0.835 for language. Higher-education value added also predicts lottery-induced changes in later choices, with pass-through estimates of 0.674 for STEM enrollment and 0.680 for institution quality. These estimates are smaller than the achievement estimates, but they are economically meaningful and statistically precise.

The heterogeneity and orthogonal-value-added results sharpen the interpretation. Achievement pass-through is stronger for students with higher baseline grade-4 math achievement, while the higher-education estimates are more stable across baseline achievement groups. In the two-dimensional model, math value added predicts math achievement but not STEM enrollment, while the STEM component orthogonal to math value added predicts STEM enrollment but not math achievement. This pattern is consistent with high schools affecting higher-education choices through dimensions not fully summarized by test-score production.

For policy, the results suggest that school quality information based only on achievement is incomplete. Test-score value added is clearly informative, but it does not exhaust the ways in which high schools shape later educational paths. A broader school-quality framework should consider both academic production and the channels through which schools affect students’ transition into higher education.

Several limitations remain. The higher-education outcomes used here capture enrollment and institution quality rather than the full application process, and future work should use the centralized application data more directly to distinguish application, admission, and enrollment margins. The value-added measures are also estimated observationally before being validated with lottery variation, so precision and shrinkage choices remain important. Finally, the orthogonal STEM results point to a distinct school dimension, but they do not by themselves identify the mechanism. Understanding whether that dimension reflects information, counseling, peer effects, curricular exposure, or preferences is the next step.

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